

ME 461 - More Pico - Less visible light

Rules of thumb: - would not hurt to remember 😊

- Your code should
 - not crash no matter what
 - include sufficient comments
 - be as lean and short as possible
 - not have self-repeating parts
 - run smoothly and be responsive
- Try to use cut to length wires. Splayed network cables are a good single strand source.
- Everytime you take your breadboard etc out from your backpack or its container, manually check connections to make sure nothing is loose or disconnected. You will get faster in time :)
- Do not modify your circuit while it is powered on
- Do not rush into anything
- If something does not work as expected:
 - it is most probably not a bug but you are missing something.
 - Re-check the cabling and make sure that there are no loose connections
- If you feel a different odor coming from around your breadboard, power it down right away, and do not directly touch things to find out what happened.
- Based on definitions in Labs / Assignments, get used to taking incentives and making reasonable assumptions when certain details are not provided.

What 2 do?

STEP 1: Get ready

Find the datasheet of QRD1114 on the internet and go over the datasheet.

The operational principle of the QRD1114 is quite simple. There is one LED emitting an IR beam (see Fig 1.c.) which then reflects back from a surface, if there is one. The reflected part of the beam falls back onto a photo-transistor. Based on the intensity of the reflection, the transistor will allow a certain amount of current to flow over it, which then can be read either as an analog or digital input as needed.

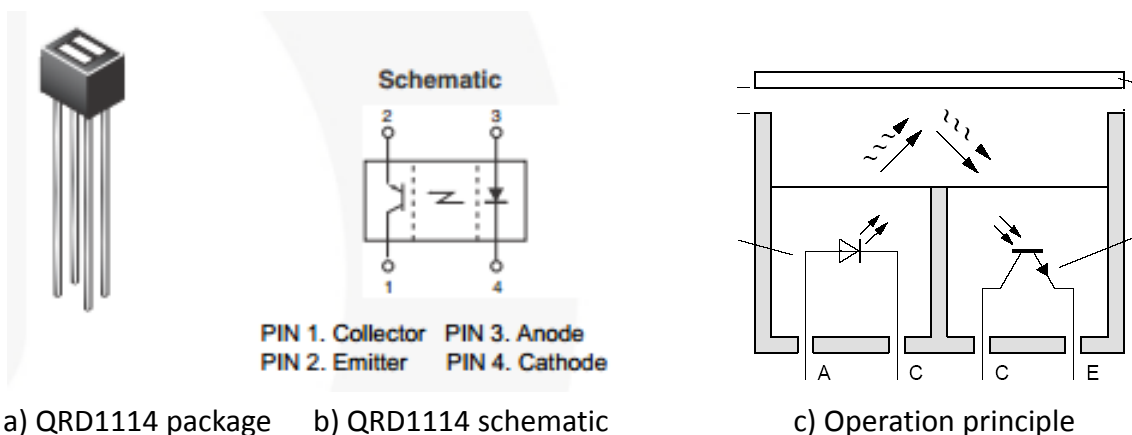


Fig 1. QRD1114: an IR transmitter-receiver pair

STEP 2: Analog input - measure distance or tell colors apart

First implement the circuit shown in Fig 2 on your breadboard. For R_1 use a low resistance (such as $100\text{-}470\Omega$ which ever available) and for R_2 use a larger resistance such as $10\text{K-}47\text{K}\Omega$. Once you are done interfacing the QRD1114 to an empty analog pin on your PICO, write a program that will print the analog VOLTAGE value at a readable pace.

Once you verify that you can read values on the PC, start experimenting with the setup.

- 1) Take an object with a flat face (a white and reflective one is preferable). In a controlled manner move it back and forth in front of the QRD1114 and observe how the reading changes.
- 2) Get a piece of white paper. Using a dark pen or a pencil, create either patterns (ex: linear or circular test pattern similar to the ones shown in Fig 3). Purposeful creative patterns are welcomed. You can also print the pattern if a printer is available. Move the test pattern at a *fixed distance* from the sensor and observe that you can detect/count the transitions between black and white regions.
- 3) Place QRD1114 at a certain distance from surfaces of different colors and observe that sensor being located at the same distance to different surfaces result in different sensor readings.
- 4) Now interface **a pot in series with R_1** , and add a **$1\text{ K}\Omega$ resistance in parallel with the pot**, and **add a $\sim 100\Omega$ resistance in parallel with R_1** (*I know you were not supposed to buy one but ...?*). Observe the effect of changing POT resistance on the detection characteristics of the QRD1114.

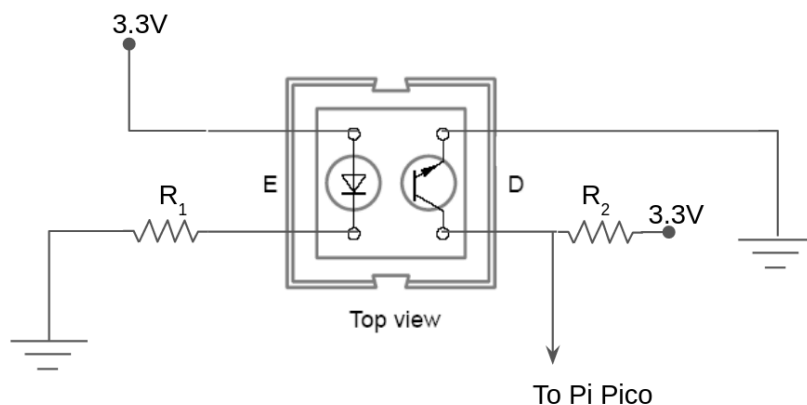


Fig 2. QRD 1114 test circuit (If you have a CNY70 the logic is similar)

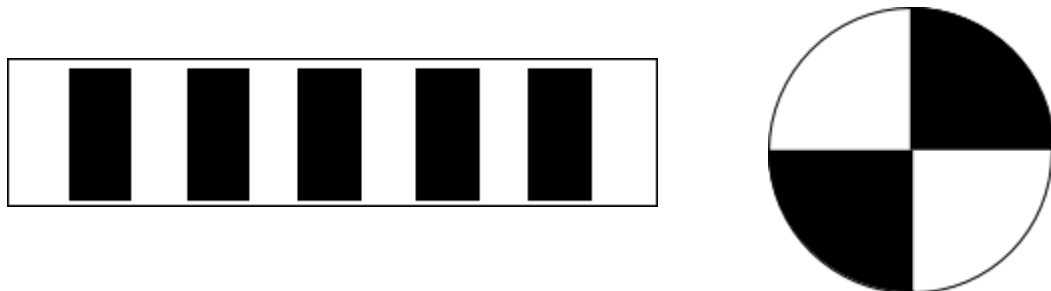


Fig 3. Sample test patterns

STEP 3: Digital input - Encoder

This time connect your signal to a digital input on your Pi Pico and detect black / white passes using an interrupt.

Step 2 and 3 are for warming up, and you are not expected to demonstrate them as long as you demonstrate STEP 4 and 5.

STEP 4: Quadrature encoder

Fig 4 illustrates the general principle of an incremental encoder. At this step you are expected to build a simple encoder from which you can obtain encoder signals as shown in Fig. 4. By pressing ButtonL, the user should be able to cycle between alternative encoding methods (i.e. X1, X2 or X4) and the selected encoding mode should be printed. Button R should reset the encoder count.

As pulses keep coming, you should increment or decrement the current count and print it.

As of now, we will **not** connect this encoder to a motor, and test it manually. Therefore, scale does not matter, indeed a slightly large one is preferred, so that we can easily see the markings on the encoder surface and correlate them with the counts you print.

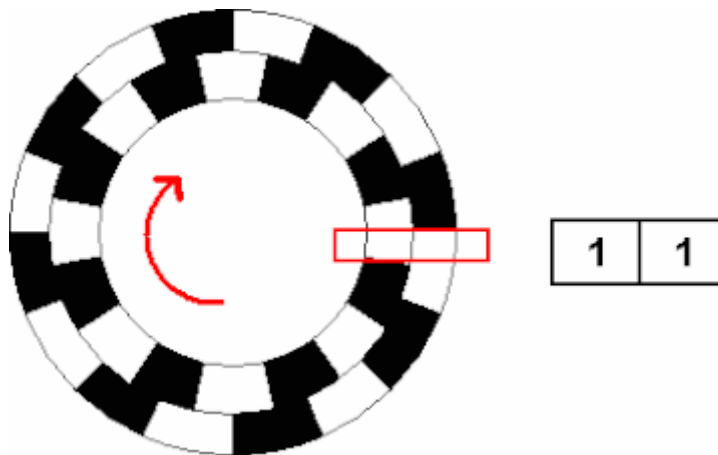


Fig 4: A Typical INCREMENTAL Encoder Schematics

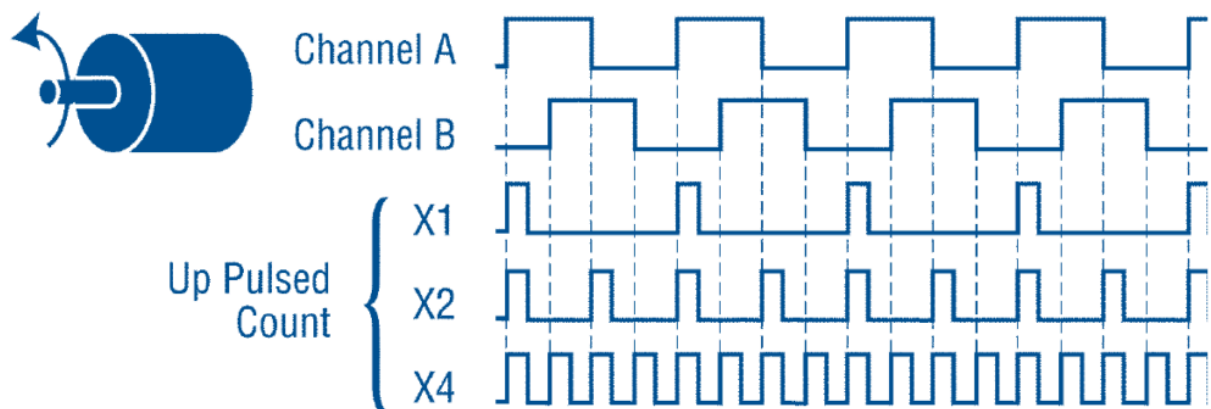


Fig 5. X1, X2 & X4 encoding

STEP 5: An analog sensor - surprise me!

Build a distance measurement sensor using QRD 1114 using the analog signal you get from it. You are also expected to document the response (i.e. the curve relating the actual distance vs your sensor's output - whatever it is that you print) and repeatability (if the distance does not change, how much your sensor output deviates), sensor resolution and if possible accuracy.

RULES of the GAME: First thing first

Have fun!

Note that your sensor is not supposed to be manufactured using conventional techniques, yet its readings should be reliable / repeatable.

Hot glue, box cutters, card boxes, old CD's, scrap materials (I like properly used scrap materials or salvaged parts) etc. are good resources.

The design is completely up to you. Try to think simple and focus on the objective!